

Vision Documentation

Interactive House - Subgroup 2: Units

Revision History.....	1
Product Overview.....	3
1. Project Background.....	3
2. Problem Statement.....	3
3. Vision Statement.....	3
4. Subgroup Goals & Scope.....	4
5. Stakeholders.....	4
6. Web Interface Perspective.....	5
7. Mobile App Perspective.....	6

Revision History

Date	Version	Description	Author
10/Feb/26	1.0	Initial Product Overview and Vision text	Ali Daoud
27/Feb/26	1.1	Revision of Vision Document	Daniel Jönsson
27/Feb/26	1.2	Web Interface Perspective	Mustafa Al-Bayati
15/Mars/26	1.3	Small Improvement in the Web interface Perspective section	Mustafa Al-Bayati
21/Apr/26	1.4	Add Table of Contents, Refactor and Add 7. Mobile Perspective	Daniel Jönsson

Product Overview

1. Project Background

The Units subgroup is responsible for developing the front-end interfaces that allow users to interact with the Interactive House. This includes creating a Web-based GUI for traditional computers and a Touch-screen interface for mobile devices.

Unlike a static app, our units must be dynamic and scalable: they will download and render User Interface (UI) components directly from the Server, which originally received them from the devices themselves.

Subgroup Responsibilities

Mobile Phone Unit: Developing a touch-based interface (comparable to Android or iPhone standards) for portable control.

Web Client: Creating a browser-based interface for laptops and desktop computers.

Alternative Interaction: Exploring advanced inputs such as speech recognition and air gestures to improve accessibility. We decided to give these implementations to SG4: Innovation.

2. Problem Statement

Modern smart homes often suffer from fragmented control systems, where each device requires its own application or interface. This creates inconsistency, confusion, and limited accessibility.

The goal of the Units subgroup is to establish a unified, centralized interaction model where any authorized unit (web or mobile) can control any registered house device through a common server interface.

For users with severe physical disabilities, the absence of a simple and centralized interface is not merely inconvenient; it represents a barrier to independence and environmental control.

3. Vision Statement

Our vision is to provide a seamless, accessible, and responsive digital "window" into the Interactive House. Through context-dependent communication and real-time synchronization with the server, users will be able to control their environment

regardless of device type (web or mobile). The system shall adapt to the user's context, ensuring accessibility whether they are seated at a desk, using a mobile device, or confined to a bed.

4. Subgroup Goals & Scope

As the Units team, our success is measured by the usability and responsiveness of the interfaces.

Successfully receiving and displaying the software components sent by the server for specific devices like a lamp ON/OFF toggle.

Ensuring users can see real-time data, such as states of the different house components/devices.

While we focus on health care and accessibility, our current implementation excludes the communication of medical data between patients and doctors, focusing strictly on house device control.

The Units subgroup consists of three developers. We work in parallel with the Server/DB and Device groups. Since we cannot and shouldn't wait for the other teams to finish, we will simulate the server environment and negotiate communication protocols with the other subgroups to ensure our interfaces work correctly during final integration.

5. Stakeholders

Our stakeholders are divided into those we report to and those we design for:

Teachers (The Customers): Within the project organization, the teachers act as the Super Project Managers and primary customers. The Units subgroup must present progress to them every two weeks and deliver a final report and presentation at the end of the course.

Persons with Disabilities (Target Users): While the general public can use the system, our primary design focus is on individuals with high grades of disabilities who may be bounded to a bed or wheelchair. Our goal is to increase their independence and self-control.

Medical Staff (Future Stakeholders): In future iterations, medical doctors and nurses could use the web-based interfaces to monitor patient health and communicate medical data.

The Project Team: The Project Manager (PM) and Requirements Manager (RM) oversee our subgroup to ensure our interfaces integrate correctly with the Server and Device groups.

6. Web Interface Perspective

From a Web development perspective, the Interactive House Web Client serves as the primary control dashboard for desktop and laptop users. The Web Interface provides a real-time dashboard for controlling smart home devices and monitoring sensor data through cloud-based synchronization.

The Web interface is designed using modern browser technologies (Next.js + Firebase) and focuses on:

- Secure user authentication
- Real-time device state synchronization
- Dynamic rendering of UI components received from the server
- Responsive layout for different screen sizes

7. Mobile App Perspective

From a Mobile development perspective, the Interactive House Mobile application serves as the portable, gesture-driven control hub for users on Android and iOS.

The Mobile interface provides a fun and great experience for interacting with smart home hardware, optimized for on-the-go monitoring and immediate sensory feedback. By leveraging React Native and NativeWind, the application bridges the gap between digital control and physical actuation with a focus on:

- Cross-Platform Performance: A unified TypeScript codebase ensuring high-performance UI rendering and consistent behavior across both iOS and Android ecosystems.
- Real-Time Hardware Interaction (music.tsx): A specialized module featuring an interactive piano simulation. This interface generates local high-fidelity audio while simultaneously communicating with the Arduino house to trigger synchronized physical responses.
- Native Sensory Feedback: Utilization of mobile-specific capabilities, such as haptics and optimized touch gestures, to make device toggling and sensor monitoring feel intuitive.
- Adaptive Styling: A fluid, responsive layout built with NativeWind that maintains design integrity across various mobile aspect ratios and screen densities.

For future work in mobile, we wish to add bluetooth capabilities, so that in case the internet goes down, the house can still be used through bluetooth.